

The Level-Six Fibre Lattice in Engel's Torus Fibration

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Abstract

Engel's complex two-torus fibration carries an integral smooth-fibre lattice $\Lambda = H_1(F, \mathbb{Z})$ with nontrivial monodromy. We determine its invariant vectors, covectors, and alternating forms in Engel's two integral markings. The invariant alternating forms constitute a rank-one lattice. With the primitive generator ξ and distinguished classes δ_0, ψ fixed as below, contraction gives

$$\iota_{\delta_0} \xi = -6\psi.$$

An explicit unimodular basis gives the normal form $J \oplus 6J$. Its discriminant pairing defines a finite Heisenberg group and an irreducible six-dimensional representation.

The topology of the logarithmic fillings requires additional integral data. We compute the complete Smith form of the two-fibre presentation, including zero determinant, and relate its determinant to the rational Seifert Euler number. For a specified effective rank-one Seifert $\mathbb{C}^*$ -bundle, the primitive integral Leray section has index $\text{lcm}(r, s)$. Its differential has coefficient $\Delta_{b_0}/\text{gcd}(r, s)$, while the global Seifert class can also contain $\mathbb{Z}/\text{gcd}(r, s)$ torsion. Engel's rank-four fibration has a separate integral transgression calculation for multiplicities $(3, 4)$ and $|p| = 1$. Comparing these differentials requires their integral sections and orientations, not just their numerical coefficients.

1 The fibre calculation

We ask which integral classes of a smooth torus fibre remain invariant under parallel transport, and what representation they determine. Let $f : Y(u) \rightarrow \mathbb{P}^1$ be the complex two-torus fibration described by Engel, and put

$$U = \mathbb{P}^1 \setminus \{0, 1, \infty\}, \quad F = f^{-1}(t), \quad \Lambda = H_1(F, \mathbb{Z}), \quad V = \Lambda^\vee$$

for $t \in U$. Over U this is also the smooth fibre lattice of the unmodified fibration $X(u) \rightarrow \mathbb{P}^1$.

Engel's construction supplies a distinguished circle class δ_0 and a primitive quotient covector ψ . They occur in the filtration

$$0 \longrightarrow \Lambda^\times \longrightarrow \Lambda \xrightarrow{\psi} \mathbb{Z} \longrightarrow 0, \quad 0 \longrightarrow \mathbb{Z}\delta_0 \longrightarrow \Lambda^\times \longrightarrow \Lambda' \longrightarrow 0, \quad (1.1)$$

where δ_0 is the circle class of the principal $\mathbb{G}_m$ -bundle in Engel's chosen marking [1, Sections 2 and 4]. Here $\Lambda^\times = \ker \psi$ and $\Lambda' = \Lambda^\times / \mathbb{Z}\delta_0$.

The monodromy also preserves a primitive alternating form ξ . The calculation below determines this form and proves

$$\iota_{\delta_0} \xi = -6\psi. \quad (1.2)$$

Thus the coefficient six relates three specified tensors on the same geometric lattice. Computing it requires the integral markings and their orientations. The resulting discriminant pairing will then define the finite Heisenberg representation.

We record the integral markings because every sign below depends on them. Let $\varrho = e^{2\pi i/3}$. In the two good-reduction markings, and writing τ_0, τ_1 for the periods of the second elliptic factors, Engel takes

$$\begin{aligned} e_1 &= (1, 0), & e_2 &= (\varrho, 0), & \delta_0 &= (0, 1), & e_4 &= (0, \tau_0), & e_3 &= \tfrac{1}{3}(2 + \varrho, 1), \\ f_1 &= (1, 0), & f_2 &= (i, 0), & \delta_1 &= (0, 1), & f_4 &= (0, \tau_1), & f_3 &= \tfrac{1}{2}(1 + i, 1). \end{aligned} \quad (1.3)$$

The first four entries in each row generate a direct-sum sublattice; after adjoining e_3 , respectively f_3 , the resulting overlattice Λ has ordered bases

$$B_0 = (e_1, e_2, e_3, e_4), \quad B_1 = (f_1, f_2, f_3, f_4).$$

For standard counterclockwise meridians $\gamma_0 \gamma_1 \gamma_\infty = 1$, the matrices act on column coordinates and are

$$[T_0]_{B_0} = \begin{pmatrix} 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad [T_1]_{B_1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.4)$$

The complex orientations give multiplication by ϱ at 0 and by $-i$ at 1. The circle classes and the transition matrix are

$$\delta_0 = -2e_1 - e_2 + 3e_3, \quad \delta_1 = -f_1 - f_2 + 2f_3, \quad C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (1.5)$$

where the columns of C express B_1 in B_0 and $C\delta_1 = \delta_0$. These are precisely the integral data of Engel's Theorem 4.1. Its proof fixes the sign at infinity by taking a simple zero, rather than a pole, of the linearization for a counterclockwise meridian [1, pp. 10–13].

In the common basis B_0 , direct conjugation and $T_0 T_1 T_\infty = I$ give

$$[T_1]_{B_0} = C[T_1]_{B_1} C^{-1} = \begin{pmatrix} -1 & 1 & -1 & 2 \\ -1 & 0 & -1 & 1 \\ 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (1.6)$$

$$[T_\infty]_{B_0} = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.7)$$

Thus $T_0^3 = T_1^4 = I$, $(T_\infty - I)^2 = 0$, and

$$(T_\infty - I)\Lambda = \langle e_1 - e_3, e_3 - e_2 \rangle. \quad (1.8)$$

Theorem 1.1 (The geometric level-six lattice). *Let $\Gamma = \langle T_0, T_1 \rangle$. For the fibre lattice and markings above,*

$$\Lambda^\Gamma = \mathbb{Z}\delta_0, \quad V^\Gamma = \mathbb{Z}\psi, \quad (\wedge^2 V)^\Gamma = \mathbb{Z}\xi, \quad (1.9)$$

where $\psi = e_4^\vee$ and

$$\xi = 3e_1^\vee \wedge e_2^\vee + e_1^\vee \wedge e_3^\vee - 2e_2^\vee \wedge e_3^\vee - 2e_3^\vee \wedge e_4^\vee. \quad (1.10)$$

With the contraction convention $(\iota_v \xi)(w) = \xi(v, w)$, identity (1.2) holds. Moreover, the ordered basis

$$a_1 = e_1, \quad b_1 = e_3, \quad a_2 = -\delta_0 = 2e_1 + e_2 - 3e_3, \quad b_2 = e_4 - 2e_1 \quad (1.11)$$

is unimodular and satisfies

$$[\xi]_{(a_1, b_1, a_2, b_2)} = J \oplus 6J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \delta_0 = -a_2, \quad \psi = b_2^\vee. \quad (1.12)$$

Consequently

$$\det(\xi^b) = 36, \quad \text{SNF}(\xi^b) = \text{diag}(1, 1, 6, 6), \quad (1.13)$$

and the elementary divisors of ξ are $(1, 6)$.

Proof. Solving $T_i v = v$ for $i = 0, 1$ gives $v = q(-2, -1, 3, 0)^T$, and solving $T_i^{-T} u = u$ gives $u = q(0, 0, 0, 1)^T$. Integrality makes the two common invariant lattices exactly $\mathbb{Z}\delta_0$ and $\mathbb{Z}\psi$.

Write the coefficients of an alternating form in the order

$$e_1^\vee \wedge e_2^\vee, e_1^\vee \wedge e_3^\vee, e_1^\vee \wedge e_4^\vee, e_2^\vee \wedge e_3^\vee, e_2^\vee \wedge e_4^\vee, e_3^\vee \wedge e_4^\vee.$$

Row reduction of $T_i^T M T_i = M$ gives the integral solutions $q(3, 1, 0, -2, 0, -2)$, proving the last equality in (1.9) and recovering Engel's invariant class from Lemma 6.2 [1, p. 15]. Its matrix in B_0 is

$$[\xi]_{B_0} = \begin{pmatrix} 0 & 3 & 1 & 0 \\ -3 & 0 & -2 & 0 \\ -1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{pmatrix}. \quad (1.14)$$

Therefore

$$\delta_0^T [\xi]_{B_0} = (0, 0, 0, -6),$$

which is (1.2).

The columns of (1.11) form the matrix

$$P = \begin{pmatrix} 1 & 0 & 2 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \det P = -1.$$

Multiplication gives $P^T [\xi]_{B_0} P = J \oplus 6J$. This proves the normal form and the Smith form. Finally,

$$\text{Pf}([\xi]_{B_0}) = -6, \quad \frac{1}{2} \xi \wedge \xi = -6e_1^\vee \wedge e_2^\vee \wedge e_3^\vee \wedge e_4^\vee,$$

so $\det[\xi]_{B_0} = (-6)^2 = 36$, as asserted. $\square$

Remark 1.2 (Basis and orientation). If the columns of $A \in \text{GL}_4(\mathbb{Z})$ are a new basis in an old one, then the simultaneous coordinate changes are

$$[\xi]' = A^T [\xi] A, \quad [\delta]' = A^{-1} [\delta], \quad [\psi]' = A^T [\psi]. \quad (1.15)$$

Thus (1.2) is tensorial. The Pfaffian changes by $\det A$, which explains the signs -6 in B_0 and $+6$ in the orientation-reversing basis (1.11), while the determinant and Smith form do not change. Reversing only δ_0 , only ψ , or the contraction convention reverses the sign in (1.2). The abstract normal form alone does not select these classes: for example, in $J \oplus 6J$, taking $\delta = a_1$ and $\psi = b_2^\vee$ gives $\iota_\delta \xi = b_1^\vee$, not -6ψ .

2 The discriminant pairing and Heisenberg module

The invariant form identifies Λ with a finite-index sublattice of its dual. The quotient measures the failure of this identification to be unimodular. Its pairing will prescribe the commutators of the finite Heisenberg group.

The following construction applies to any rank-four alternating lattice (L, ξ) with normal form $J \oplus 6J$ in a basis (a_1, b_1, a_2, b_2) . For Engel's fibration, take $L = \Lambda$ and the basis in Theorem 1.1. Define

$$\xi^\flat : L \longrightarrow L^\vee, \quad \xi^\flat(x) = \xi(x, -), \quad \mathcal{D}_\xi = L^\vee / \xi^\flat(L).$$

The rational extension $\phi = \xi^\flat \otimes \mathbb{Q}$ is an isomorphism.

Lemma 2.1 (Discriminant pairing). *For $u, v \in L^\vee$, the formula*

$$b_\xi(\bar{u}, \bar{v}) = u(\phi^{-1}v) \pmod{\mathbb{Z}} \quad (2.1)$$

defines a perfect alternating pairing $b_\xi : \mathcal{D}_\xi \times \mathcal{D}_\xi \rightarrow \mathbb{Q}/\mathbb{Z}$. Moreover, if

$$\alpha = [a_2^\vee], \quad \beta = [b_2^\vee],$$

then

$$\mathcal{D}_\xi = \langle \alpha \rangle \oplus \langle \beta \rangle \cong (\mathbb{Z}/6\mathbb{Z})^2, \quad b_\xi(\alpha, \beta) = \frac{1}{6}, \quad b_\xi(\beta, \alpha) = -\frac{1}{6}. \quad (2.2)$$

Proof. Write $x = \phi^{-1}u$ and $y = \phi^{-1}v$. Replacing u by $u + \xi^\flat(\lambda)$ changes (2.1) by $\xi(\lambda, y) = -v(\lambda) \in \mathbb{Z}$; replacing v by $v + \xi^\flat(\mu)$ changes it by $u(\mu) \in \mathbb{Z}$. The pairing is therefore well defined and alternating. If it annihilates $\bar{u}$ against every $\bar{v}$, then $v(x) = -u(\phi^{-1}v)$ is integral for every $v \in L^\vee$. Hence $x \in L$, so $\bar{u} = 0$. The induced homomorphism to the Pontryagin dual is injective between finite groups of the same order, and is therefore an isomorphism. Formula (2.2) follows directly from the normal form. $\square$

Exponentiating b_ξ gives a μ_6 -valued alternating pairing. We realize it as the commutator of a central extension of $\mathcal{D}_\xi$. Choosing the two cyclic summands in (2.2) then gives translation and multiplication operators on functions on one summand. Their commutation relation will recover the original discriminant pairing.

Put $K = \langle \alpha \rangle \cong \mathbb{Z}/6\mathbb{Z}$, let $\zeta_6 = e^{2\pi i/6}$, and let $\chi \in \widehat{K}$ satisfy $\chi(\alpha) = \zeta_6$. The map

$$r\alpha + s\beta \longmapsto (r\alpha, \chi^s)$$

is a polarization $\mathcal{D}_\xi \cong K \oplus \widehat{K}$. Define

$$H_6 = \mu_6 \times K \times \widehat{K}$$

with multiplication

$$(z, u, \eta)(z', u', \eta') = (zz'\eta'(u), u + u', \eta\eta'). \quad (2.3)$$

Theorem 2.2 (Level-six Heisenberg module). *The center of H_6 is the displayed copy of μ_6 . Its commutator pairing is*

$$[(1, u, \eta), (1, u', \eta')] = (\eta'(u)\eta(u')^{-1}, 0, 1) = (e^{2\pi i b_\xi(d, d')}, 0, 1), \quad (2.4)$$

where $d \leftrightarrow (u, \eta)$ and $d' \leftrightarrow (u', \eta')$. On the six-dimensional space $\mathcal{H} = \mathbb{C}[K]$, the formula

$$[\rho(z, u, \eta)h](t) = z\eta(t)h(t + u) \quad (2.5)$$

defines an irreducible representation, and the central μ_6 acts by its faithful tautological character.

Proof. The bicharacter in (2.3) makes the multiplication associative, and direct multiplication gives (2.4). Its nondegeneracy forces a central element to have $u = 0$ and $\eta = 1$. The same calculation verifies (2.5).

For $t \in K$, let δ_t be the delta-function at t . The operator

$$P_t = \frac{1}{6} \sum_{\eta \in \widehat{K}} \eta(t)^{-1} \rho(1, 0, \eta)$$

projects onto $\mathbb{C}\delta_t$. A nonzero invariant subspace therefore contains some δ_t , and translations carry it through every delta-function. These span $\mathcal{H}$, proving irreducibility. $\square$

Applied to Theorem 1.1, this construction belongs to the fibre lattice $H_1(F, \mathbb{Z})$. It is not constructed from the ordinary cohomology of the total six-manifold.

3 Two multiple fibres

The discriminant group describes the integral pairing within a smooth fibre. The topology of the total space also depends on how fibres are filled. Two logarithmic fillings lead to relations involving a regular circle class and two meridians. A global section integer records their product relation.

We first compute the resulting abstract group for all integral data. We then identify the effective Seifert cases and their Euler number. Engel's geometric presentation will enter in Section 5 through its vanishing cycles and logarithmic fillings.

Theorem 3.1 (Two-fibre Smith form). *Let $r, s > 0$, let $m, n, b_0 \in \mathbb{Z}$, and define*

$$G_{b_0} = \left\langle c, x_0, x_1 \mid c \text{ central, } x_0 x_1 = c^{b_0}, x_0^r = c^m, x_1^s = c^n \right\rangle. \quad (3.1)$$

Put

$$\Delta_{b_0} = rn + sm - rsb_0, \quad d = \gcd(r, s, m, n). \quad (3.2)$$

Then

$$G_{b_0} \cong \text{coker} \begin{pmatrix} r & s \\ -m & n - sb_0 \end{pmatrix}. \quad (3.3)$$

If $\Delta_{b_0} \neq 0$, then

$$G_{b_0} \cong \mathbb{Z}/d\mathbb{Z} \oplus \mathbb{Z}/(|\Delta_{b_0}|/d)\mathbb{Z}, \quad |G_{b_0}| = |\Delta_{b_0}|. \quad (3.4)$$

If $\Delta_{b_0} = 0$, then

$$G_{b_0} \cong \mathbb{Z} \oplus \mathbb{Z}/d\mathbb{Z}. \quad (3.5)$$

In either case G_{b_0} is cyclic if and only if $d = 1$.

Proof. The product relation gives $x_1 = x_0^{-1}c^{b_0}$, so the group is abelian. In additive notation its two remaining relation vectors in $\mathbb{Z}x_0 \oplus \mathbb{Z}c$ are

$$rx_0 - mc, \quad sx_0 + (n - sb_0)c,$$

which proves (3.3). The greatest common divisor of the entries is

$$\gcd(r, s, m, n - sb_0) = \gcd(r, s, m, n) = d,$$

and the determinant is Δ_{b_0} . For nonzero determinant, the Smith invariants are d and $|\Delta_{b_0}|/d$. If the determinant vanishes, the matrix still has rank one because $r, s > 0$, so its Smith form is $\text{diag}(d, 0)$. The cyclicity assertions follow. $\square$

Proposition 3.2 (Section gauges and orientations). *For $k_0, k_1 \in \mathbb{Z}$, the generator change*

$$x'_0 = x_0 c^{k_0}, \quad x'_1 = x_1 c^{k_1} \quad (3.6)$$

sends

$$m' = m + rk_0, \quad n' = n + sk_1, \quad b'_0 = b_0 + k_0 + k_1 \quad (3.7)$$

and preserves both d and Δ_{b_0} . In the gauge $b_0 = 0$, the presentation-preserving choice $k_1 = -k_0 = -k$ sends

$$(m, n) \mapsto (m + rk, n - sk). \quad (3.8)$$

Interchanging (r, m, x_0) with (s, n, x_1) preserves Δ_{b_0} . Replacing the oriented fibre generator by $c' = c^{-1}$ sends (m, n, b_0) to $(-m, -n, -b_0)$ and sends Δ_{b_0} to $-\Delta_{b_0}$.

Proof. Substitution in the three defining relations gives (3.7), and

$$r(n + sk_1) + s(m + rk_0) - rs(b_0 + k_0 + k_1) = \Delta_{b_0}.$$

The remaining statements follow by the indicated changes of generators. They also show why only $|\Delta_{b_0}|$ is intrinsic without fixed orientations. $\square$

The hypotheses in Theorem 3.1 cannot be weakened silently. For $b_0 = 0$,

$$(r, s; m, n) = (2, 4; 2, 2) \implies G_0 \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z},$$

although $\Delta_0 = 12$. Also

$$(2, 4; 2, -4) \implies G_0 \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, \quad \Delta_0 = 0.$$

At the simplest zero-coefficient boundary, $m = n = 0$ gives $G_0 \cong \mathbb{Z} \oplus \mathbb{Z}/\gcd(r, s)\mathbb{Z}$.

Proposition 3.3 (Seifert Euler number). *Let M be an oriented Seifert manifold over S^2 with two normalized effective exceptional fibres, written in Wilkes's convention as*

$$(b_0, S^2; (r, \beta_0), (s, \beta_1)), \quad (3.9)$$

where $r, s \geq 2$, $0 < \beta_0 < r$, $0 < \beta_1 < s$, and $\gcd(r, \beta_0) = \gcd(s, \beta_1) = 1$. With

$$m = -\beta_0, \quad n = -\beta_1, \quad (3.10)$$

one has $\pi_1(M) \cong G_{b_0}$ and

$$e(M) = -\left(b_0 + \frac{\beta_0}{r} + \frac{\beta_1}{s}\right) = -b_0 + \frac{m}{r} + \frac{n}{s}, \quad rs e(M) = \Delta_{b_0}. \quad (3.11)$$

In particular $d = 1$, and hence

$$\pi_1(M) \cong \begin{cases} \mathbb{Z}/|\Delta_{b_0}|\mathbb{Z}, & e(M) \neq 0, \\ \mathbb{Z}, & e(M) = 0. \end{cases} \quad (3.12)$$

If $g = \gcd(r, s)$ and $L = \text{lcm}(r, s) = rs/g$, then the smaller denominator clearing is

$$Le(M) = -Lb_0 + \frac{s}{g}m + \frac{r}{g}n = \frac{\Delta_{b_0}}{g} \in \mathbb{Z}. \quad (3.13)$$

Proof. Wilkes's presentation has central regular fibre h , local relations $a_i^{\alpha_i} h^{\beta_i} = 1$, and, over S^2 with two exceptional fibres, the product relation $a_0 a_1 = h^{b_0}$. Setting $c = h$, $x_i = a_i$ gives (3.1) and (3.10). His Euler-number convention gives (3.11) [4, Proposition 4.3, p. 149]. Effectiveness implies $d = 1$, and Theorem 3.1 gives (3.12). Equation (3.13) is immediate. Hatcher's section-change calculation gives the geometric counterpart of (3.6): twists are added at one boundary torus and subtracted at another, leaving the Euler sum fixed [2, Section 2.1, pp. 25–27]. $\square$

Remark 3.4. The rational number $e(M)$ is orientation-sensitive. Reversing the fibre orientation and renormalizing Wilkes's two local pairs sends

$$(b_0; \beta_0, \beta_1) \mapsto (-b_0 - 2; r - \beta_0, s - \beta_1),$$

which negates both $e(M)$ and Δ_{b_0} but preserves the abstract group. With exceptional fibres, $e(M)$ is a rational Seifert Euler number; only in the ordinary circle-bundle case is it itself the integral Euler class on the underlying base.

4 The rank-one Seifert transgression

The rational Seifert Euler number determines a Leray differential only after the integral fibre section has been specified. Multiple fibres restrict which rational sections extend integrally over the base. We compute this restriction for an effective Seifert $\mathbb{C}^*$ -bundle with two multiple fibres.

Let $r, s \geq 2$, let $0 < \beta_0 < r$ and $0 < \beta_1 < s$, and assume

$$\gcd(r, \beta_0) = \gcd(s, \beta_1) = 1. \quad (4.1)$$

For $b_0 \in \mathbb{Z}$, consider Kollár's effective Seifert $\mathbb{C}^*$ -bundle

$$q : Z = Y\left(\mathcal{O}_{\mathbb{P}^1}(b_0), \frac{\beta_0}{r}[0] + \frac{\beta_1}{s}[1]\right) \longrightarrow \mathbb{P}^1. \quad (4.2)$$

Equivalently, this is the coarse map of the complement of the zero section in the corresponding line bundle on the labelled root orbisphere. The local characters are faithful by (4.1), so r and s are the actual fibre multiplicities.

Give the base its complex orientation and put $\omega = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$. Kollár's natural trivialization

$$\tau : R^1 q_* \mathbb{Q} \xrightarrow{\sim} \mathbb{Q}_{\mathbb{P}^1}$$

defines the normalized rational fibre class $u_K = \tau^{-1}(1)$. We use the Wilkes-oriented class $u = -u_K$ and set

$$m = -\beta_0, \quad n = -\beta_1, \quad e = -b_0 + \frac{m}{r} + \frac{n}{s}, \quad \Delta_{b_0} = rn + sm - rsb_0. \quad (4.3)$$

Put

$$g = \gcd(r, s), \quad \ell = \text{lcm}(r, s) = \frac{rs}{g}.$$

Theorem 4.1 (Coarse two-fibre transgression). *For the map (4.2), one has $R^0 q_* \mathbb{Z} \cong \mathbb{Z}_{\mathbb{P}^1}$ and an exact sequence of constructible sheaves*

$$0 \longrightarrow R^1 q_* \mathbb{Z} \longrightarrow \mathbb{Z}_{\mathbb{P}^1} \longrightarrow (\mathbb{Z}/r\mathbb{Z})_{\{0\}} \oplus (\mathbb{Z}/s\mathbb{Z})_{\{1\}} \longrightarrow 0, \quad (4.4)$$

where the last map sends 1 to $(1, 1)$. Consequently

$$H^0(\mathbb{P}^1, R^1 q_* \mathbb{Z}) = \ell \mathbb{Z} u, \quad H^1(\mathbb{P}^1, R^1 q_* \mathbb{Z}) \cong \mathbb{Z}/g\mathbb{Z}. \quad (4.5)$$

In the integral Leray spectral sequence of q , the differential

$$d_2^{0,1} : H^0(\mathbb{P}^1, R^1 q_* \mathbb{Z}) \longrightarrow H^2(\mathbb{P}^1, R^0 q_* \mathbb{Z}) \quad (4.6)$$

satisfies

$$d_2^{0,1}(\ell u) = \frac{\Delta_{b_0}}{g} \omega, \quad d_2^{0,1}(rsu) = \Delta_{b_0} \omega. \quad (4.7)$$

The first equation uses the primitive global section. The second uses its g -fold multiple.

Proof. Kollár's multiplicity criterion gives the two multiplicities r and s . His integral comparison map and its local cokernel give (4.4) [3, Theorem 7 and Propositions 43 and 47]. The long exact cohomology sequence of (4.4) begins with the diagonal reduction map

$$\mathbb{Z} \longrightarrow \mathbb{Z}/r\mathbb{Z} \oplus \mathbb{Z}/s\mathbb{Z}.$$

Its kernel is $\ell \mathbb{Z}$ and its cokernel is $\mathbb{Z}/g\mathbb{Z}$, proving (4.5).

In Kollár's convention the rational Chern class is

$$c_1^K(Z/\mathbb{P}^1) = \left(b_0 + \frac{\beta_0}{r} + \frac{\beta_1}{s} \right) \omega = -e\omega. \quad (4.8)$$

The quotient $Z \rightarrow Z/\mu_\ell$ induces a map of Leray spectral sequences, and Z/μ_ℓ is an ordinary $\mathbb{C}^*$ -bundle whose Chern class is ℓc_1^K . Hence

$$d_2^{0,1}(\ell u_K) = \ell c_1^K(Z/\mathbb{P}^1)$$

[3, Definition 9, Map 13, and Corollary 44]. Replacing u_K by $u = -u_K$ and using

$$\ell e = -\ell b_0 + \frac{s}{g} m + \frac{r}{g} n = \frac{\Delta_{b_0}}{g}$$

proves the first equation in (4.7). Since $rs = g\ell$, the second follows. $\square$

The coarse coefficient records only the free part of the global Seifert class. To see the missing information integrally, let

$$\mathcal{S}_{r,s} = \langle H, X_0, X_1 \mid rX_0 = H, sX_1 = H \rangle, \quad (4.9)$$

the line-bundle class group of the labelled root orbisphere. Here H is the pullback of the positive coarse class and X_0, X_1 are the two root classes. In the orientation of (4.3), the Seifert class is

$$\varepsilon = -b_0H + mX_0 + nX_1. \quad (4.10)$$

Proposition 4.2 (The global class and its coarse shadow). *There is an isomorphism*

$$\mathcal{S}_{r,s} \cong \mathbb{Z} \oplus \mathbb{Z}/g\mathbb{Z}. \quad (4.11)$$

The section gauges (3.7) leave ε fixed, and

$$\ell\varepsilon = \frac{\Delta_{b_0}}{g}H. \quad (4.12)$$

Equivalently, on gauge classes

$$\kappa[b_0, m, n] = -\ell b_0 + \frac{\ell}{r}m + \frac{\ell}{s}n = \frac{\Delta_{b_0}}{g} \quad (4.13)$$

is a surjection to $\mathbb{Z}$ whose kernel is the torsion subgroup $\mathbb{Z}/g\mathbb{Z}$. Thus the rational Euler number, and hence the coarse differential, determines the full global class exactly when $g = 1$.

Proof. Kollár's classification gives the representatives (b_0, m, n) , while a change of local section gives the relations

$$(b_0, m, n) \sim (b_0 + k_0 + k_1, m + rk_0, n + sk_1).$$

The first determinantal divisor of the relation matrix is 1, while the second is

$$\gcd(r, s, rs) = g.$$

Its Smith form therefore gives (4.11). Relations $rX_0 = H$ and $sX_1 = H$ prove both gauge invariance and (4.12). The three coefficients in (4.13) have greatest common divisor 1, so κ is onto. Its finite kernel has order g , proving the last assertion. $\square$

The torsion is already present in the smallest rank-drop example. For

$$r = s = 2, \quad b_0 = -1, \quad \beta_0 = \beta_1 = 1, \quad (4.14)$$

one has $e = \Delta_{b_0} = 0$, so the coarse transgression vanishes, whereas

$$\varepsilon = H - X_0 - X_1 = X_0 - X_1$$

is the nonzero element of order two in $\mathcal{S}_{2,2}$. Thus a zero rational Euler number need not mean that the global Seifert class is zero.

Proposition 4.3 (Effectivization and rank drop). *For formal local data (r, m) and (s, n) , put*

$$a_0 = \gcd(r, m), \quad a_1 = \gcd(s, n), \quad R = \frac{r}{a_0}, \quad S = \frac{s}{a_1}, \quad h = \gcd(R, S). \quad (4.15)$$

If the data arise from a rank-one Seifert construction after reducing the two local fractions, then the actual local orders are R, S , the primitive carrier is

$$Q = \text{lcm}(R, S) = \frac{rs}{a_0 a_1 h}, \quad (4.16)$$

and

$$d_2^{0,1}(Qu) = \frac{\Delta_{b_0}}{a_0 a_1 h} \omega, \quad d_2^{0,1}(rsu) = \Delta_{b_0} \omega. \quad (4.17)$$

If the local pairs are effective and $\Delta_{b_0} = 0$, then

$$r = s, \quad m + n = r b_0. \quad (4.18)$$

For normalized pairs this says $b_0 = -1$ and $\beta_0 + \beta_1 = r$; in this case $G_{b_0} \cong \mathbb{Z}$.

Proof. The least positive integer k for which $r \mid km$ and $s \mid kn$ is $\text{lcm}(R, S)$, proving (4.16). Writing $m = a_0 M$ and $n = a_1 N$ gives

$$Qe = \frac{\Delta_{b_0}}{a_0 a_1 h}, \quad rs = (a_0 a_1 h)Q,$$

which proves (4.17). If the pairs are effective and $\Delta_{b_0} = 0$, then reduction modulo r gives $r \mid s$, and reduction modulo s gives $s \mid r$. This proves (4.18). The normalized conclusion follows from $0 < \beta_i < r$, and Theorem 3.1 gives the last assertion. $\square$

Remark 4.4 (Naturality, orientations, and scope). The naturality in Theorem 4.1 is the map of Leray spectral sequences induced by $Z \rightarrow Z/\mu_\ell$; the pullback of the ordinary fibre generator is ℓu_K . More generally, a morphism of oriented Seifert bundles gives commuting Leray differentials. If a base pullback has actual carrier $\ell' u'$ with $\ell' \mid \ell$, then the pullback of ℓu is $(\ell/\ell')\ell' u'$, so primitive numerical coefficients must be compared with this index. Reversing the fibre generator or the chosen generator ω reverses the displayed scalar. Section gauges and interchange of the two labelled points preserve (4.7); the only changes of the oriented rank-one fibre basis are $u \mapsto \pm u$.

This theorem is specific to the rank-one Seifert bundle (4.2). An affine torus fibration has a higher-rank local system, and local multiplicities and a group presentation do not determine its extension across singular fibres or its rational transgression. Transporting (4.7) to such a fibration would require an actual map over the base inducing the claimed morphism of constructible sheaves. No such map is used below.

5 The Leray carrier in Engel's (3,4) case

For Engel's fibration, the invariant covector ψ gives a section of the rank-four cohomology local system over U . Computing its transgression first requires determining which multiples extend across the singular fibres. This extension depends on the logarithmic transforms and the cusp

gluing of the actual map $f : Y(u) \rightarrow \mathbb{P}^1$. We specify these data before applying Engel's integral calculation.

Choose lifts of Engel's logarithmic-transform classes

$$a_0 \in \frac{1}{3}\Lambda^{T_0}, \quad a_1 \in \frac{1}{4}\Lambda^{T_1},$$

and put

$$v_0 = 3a_0, \quad v_1 = 4a_1, \quad m = \psi(v_0), \quad n = \psi(v_1), \quad p = 12\psi(a_0 + a_1) = 4m + 3n. \quad (5.1)$$

When the classes of a_0, a_1 are primitive torsion classes, the fibres at 0 and 1 have multiplicities 3 and 4, respectively [1, Corollary 2.10].

The carrier of the transgression is the cohomological Leray spectral sequence of the specific map $f : Y(u) \rightarrow \mathbb{P}^1$,

$$E_2^{a,b} = H^a(\mathbb{P}^1, R^b f_* \mathbb{Z}) \implies H^{a+b}(Y(u), \mathbb{Z}). \quad (5.2)$$

Over U , the restriction of $R^1 f_* \mathbb{Z}$ is a local system with fibre $V = H^1(F, \mathbb{Z})$ and monodromy T_i^{-T} . Across the three singular values, $R^1 f_* \mathbb{Z}$ is the corresponding constructible sheaf, not a local system. The fibres of f are connected, so $R^0 f_* \mathbb{Z} \cong \mathbb{Z}_{\mathbb{P}^1}$. Thus the first relevant differential is precisely

$$d_2^{0,1} : H^0(\mathbb{P}^1, R^1 f_* \mathbb{Z}) \longrightarrow H^2(\mathbb{P}^1, R^0 f_* \mathbb{Z}). \quad (5.3)$$

Theorem 5.1 (Engel's integral transgression). *Assume $|p| = 1$, as in Engel's integral-homology calculation, and choose the common orientation for which $H^2(\mathbb{P}^1, R^0 f_* \mathbb{Z}) = \mathbb{Z}\omega$. Then*

$$H^0(\mathbb{P}^1, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\psi, \quad d_2^{0,1}(12\psi) = p\omega. \quad (5.4)$$

Proof. The extension lattice in (5.4) is Engel's Lemma 6.5. For the differential, the two local affine lifts satisfy

$$\tilde{\rho}_0^3 = t_{v_0}, \quad \tilde{\rho}_1^4 = t_{v_1}.$$

Their oriented contributions to the obstruction for 12ψ are

$$\frac{12\psi(v_0)}{3} = 4m, \quad \frac{12\psi(v_1)}{4} = 3n. \quad (5.5)$$

The cusp gluing at infinity contributes zero in this marking. Their sum is $p = 4m + 3n$, which is Engel's Lemma 6.6 [1, pp. 17–18]. Reversing either generator in (5.4) reverses the corresponding sign; simultaneous transport of the orientation preserves the identity. $\square$

Proposition 5.2 (The (3, 4) presentation). *For the same geometric data,*

$$\pi_1(Y(u)) \cong \langle c, x_0, x_1 \mid c \text{ central}, x_0 x_1 = 1, x_0^3 = c^m, x_1^4 = c^n \rangle. \quad (5.6)$$

Consequently

$$\pi_1(Y(u)) \cong \begin{cases} \mathbb{Z}/|p|\mathbb{Z}, & p \neq 0, \\ \mathbb{Z}, & p = 0. \end{cases} \quad (5.7)$$

In particular it is trivial when $|p| = 1$.

Proof. Engel's vanishing cycles at infinity kill $\ker \psi$ and leave the central class c represented by e_4 . The two logarithmic fillings give (5.6) [1, Proposition 5.1, pp. 13–14]. This is Theorem 3.1 with $(r, s; b_0) = (3, 4; 0)$, for which $d = 1$ and $\Delta_0 = p$. $\square$

Comparison with the rank-one theorem

The two transgressions can be compared numerically after normalizing their integral data. For Engel's raw data $(r, s; b_0) = (3, 4; 0)$, the determinant is $\Delta_0 = 4m + 3n = p$. Under the hypothesis $|p| = 1$, reduction modulo 3 and 4 gives

$$\gcd(3, m) = \gcd(4, n) = 1.$$

Choose normalized residues

$$0 < \beta_0 < 3, \quad 0 < \beta_1 < 4, \quad -\beta_0 \equiv m \pmod{3}, \quad -\beta_1 \equiv n \pmod{4}.$$

They define faithful local characters. Put

$$k_0 = \frac{-\beta_0 - m}{3}, \quad k_1 = \frac{-\beta_1 - n}{4}, \quad b'_0 = k_0 + k_1.$$

The section gauges (3.7) replace $(0, m, n)$ by $(b'_0, -\beta_0, -\beta_1)$ and preserve the determinant. These normalized data therefore define an effective rank-one bundle of the form (4.2), with $\Delta_{b'_0} = p$. Since $g = 1$ and $\ell = 12$, its differential is

$$d_2^{0,1}(12u) = p\omega.$$

Engel's differential (5.4) has the same coefficient after the orientations are matched. Its source, however, comes from a different constructible sheaf. Engel's sheaf has generic rank four, nontrivial monodromy, and an additional singular value at infinity. A geometric comparison would require a map over the base inducing a morphism of the integral coefficient sheaves and compatible Leray differentials. On global sections it would have to compare $12u$ with 12ψ , including their extension across every singular fibre. Such a comparison is not constructed by the normalization above. Engel's Lemmas 6.5–6.6 supply the carrier and differential in (5.4).

The broader transgression problem for an affine torus fibration requires its full local system, specialization maps, global affine or Chern class, and orientations. The group presentation (3.1) supplies none of these sheaf maps by itself. Constructing their comparison with the rank-one bundle remains a geometric problem even when the numerical differentials agree.

Finally, $\Lambda = H_1(F, \mathbb{Z})$ is an auxiliary fibre lattice. In the case in which Engel's Theorem 3.1 identifies the total space with S^6 , one has

$$H^j(Y(u), \mathbb{Z}) = 0 \quad (0 < j < 6),$$

so neither Λ nor its discriminant group is ordinary positive-degree cohomology of the total space.

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